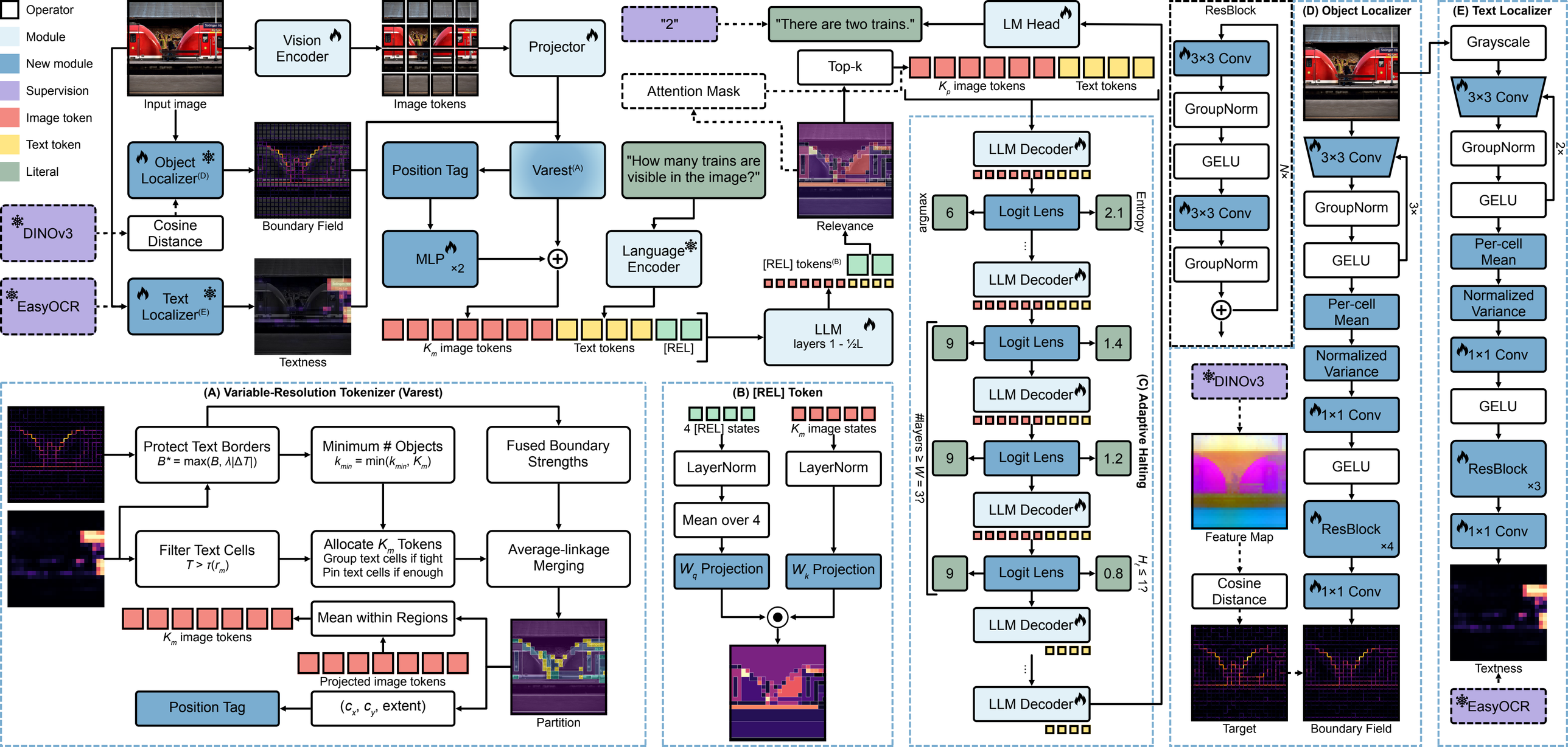

Operator
Module
New module
Supervision
Image token
Text token
Literal
Input image
Vision Encoder
Image tokens
Projector
"2"
"There are two trains."
LM Head
Top-k
Attention Mask
K_p image tokens
Text tokens
Object Localizer(D)
Boundary Field
Position Tag
Varest(A)
"How many trains are visible in the image?"
Relevance
DINOv3
Cosine Distance
MLP ×2
Language Encoder
[REL] tokens(B)
EasyOCR
Text Localizer(E)
Textness
K_m image tokens
Text tokens
[REL]
LLM layers 1 - ½L
LLM Decoder
argmax
Logit Lens
Entropy
#layers ≥ W = 3?
$H_i \leq 1$?
(C) Adaptive Halting
ResBlock
3×3 Conv
GroupNorm
GELU
×N
(D) Object Localizer
Feature Map
Target
(E) Text Localizer
Grayscale
2×
3×
Per-cell Mean
Normalized Variance
1×1 Conv
ResBlock ×3
ResBlock ×4
(A) Variable-Resolution Tokenizer (Varest)
Protect Text Borders
$B^* = \max(B, \lambda|\Delta T|)$
Minimum # Objects
$k_{min} = \min(k_{min}, K_m)$
Fused Boundary Strengths
Filter Text Cells
$T > \tau(r_m)$
Allocate K_m Tokens
Group text cells if tight
Pin text cells if enough
Average-linkage Merging
Mean within Regions
Projected image tokens
Position Tag
$(c_x, c_y, \text{extent})$
Partition
(B) [REL] Token
4 [REL] states
K_m image states
LayerNorm
Mean over 4
W_q Projection
W_k Projection